Generative AI

Introduction

KI ist so grundlegend wie Elektrizität. Sie wird alles verändern – wie wir leben, arbeiten und spielen.

Jeff Bezos

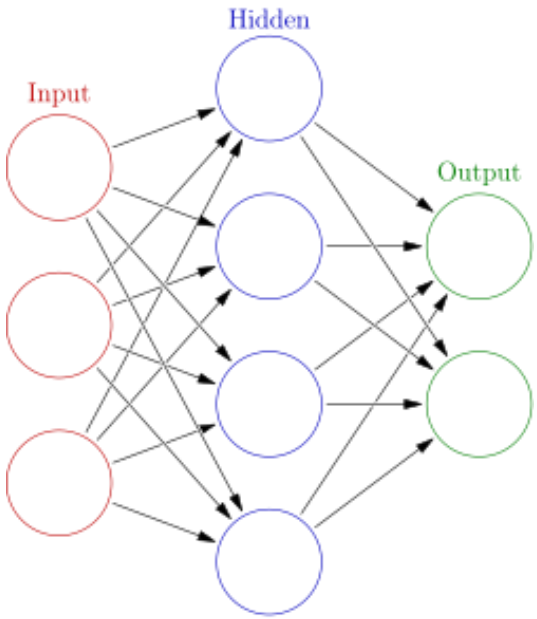
Generative KI kann als eine der kreativsten Kräfte in den kommenden Jahrzehnten betrachtet werden, mit dem Potential, in jedem Bereich des menschlichen Unterfangens Innovationen zu fördern.

Demis Hassabis

Generative AI

How do Artificial Neural Networks work?

- Inspired by human brain's neural connections
- Composed of interconnected nodes (called neurons)
- Each connection has a weight corresponding to its importance
- Hidden Layer(s)
  - Perform calculations on inputs using weighted connections
  - Each neuron applies activation function
  - Multiple hidden layers capture complex patterns
- Training process adjusts connection weights based on data



Receives input data

Has neurons and connections

Returns outputs

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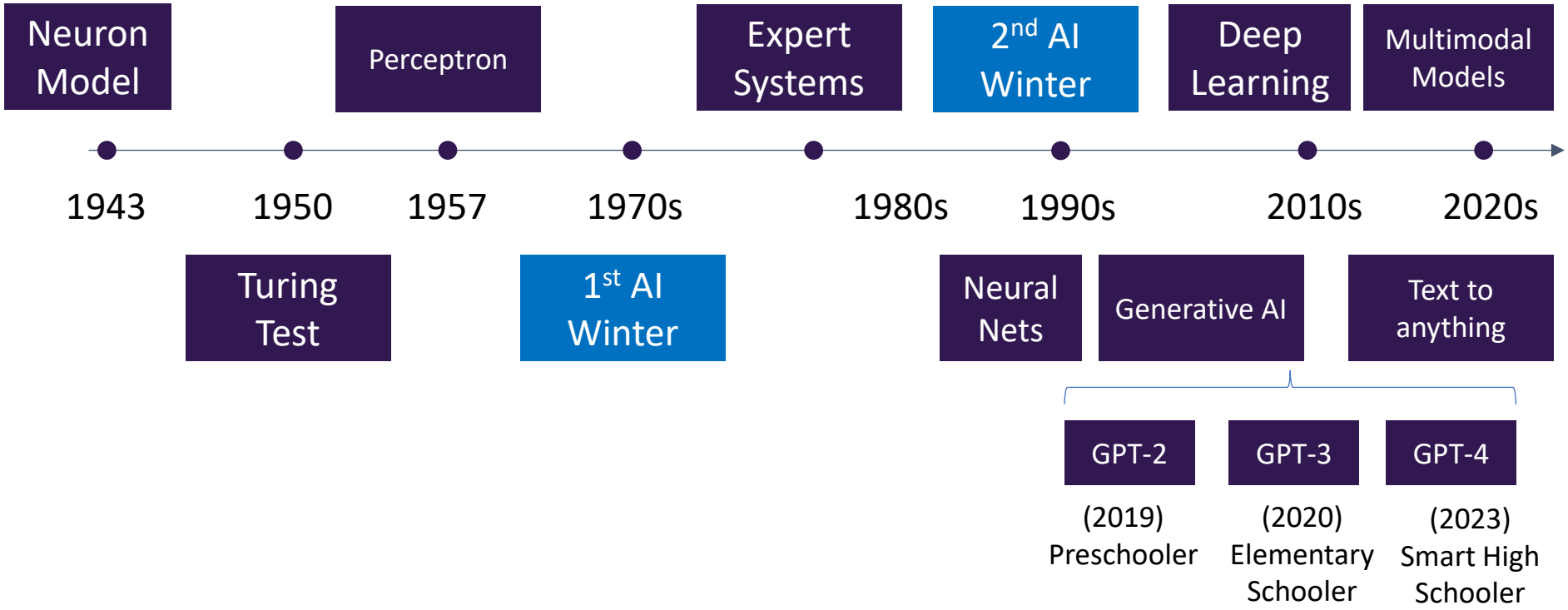
Artificial Intelligence over Time: Past and Present



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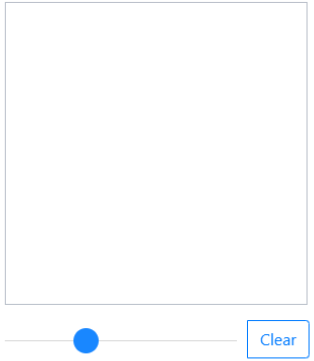
Artificial Intelligence over Time: Past and Present

SKETCHER

Sketcher is a simple (but still impressive) tool -built by Zaid Alyafeai- that uses CNN to recognize drawings. The CNN was trained to recognize 100 classes using the quick draw dataset. It achieves 6% top 5 error. The accuracy could be improved but I focused on making the model light with around 1 million parameters. The table will show the top 5 predictions.

—

Model Loaded



https://www.dar-analytics.de/sketcher.html

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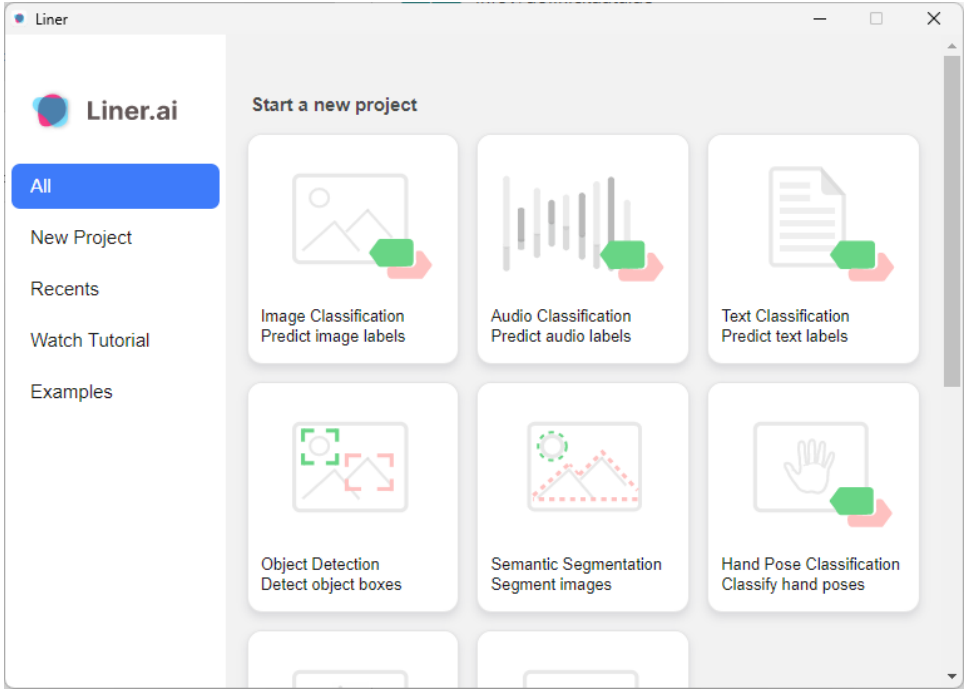
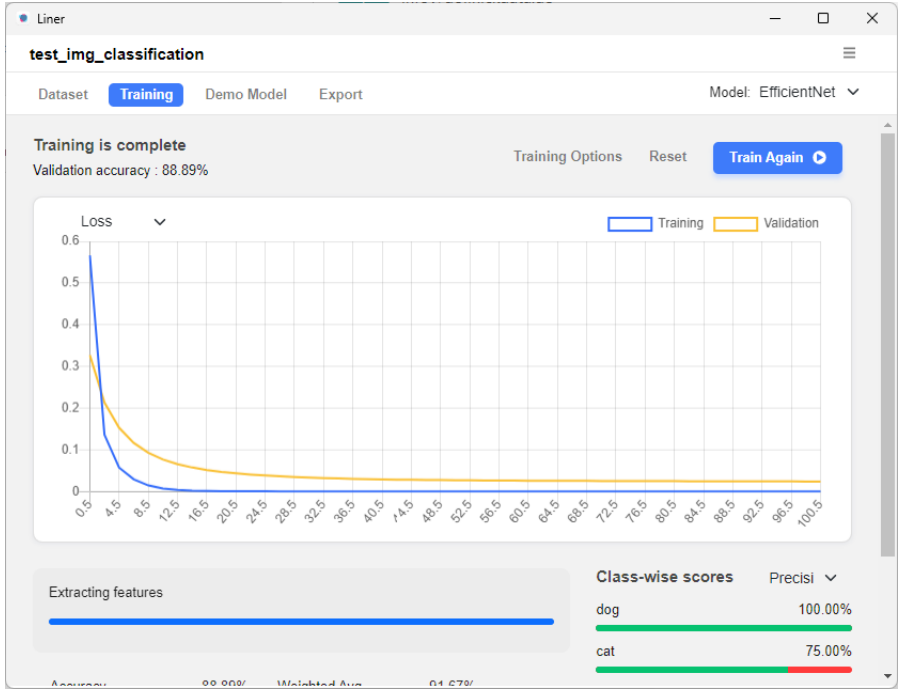
Artificial Intelligence over Time: Past and Present





https://www.liner.ai

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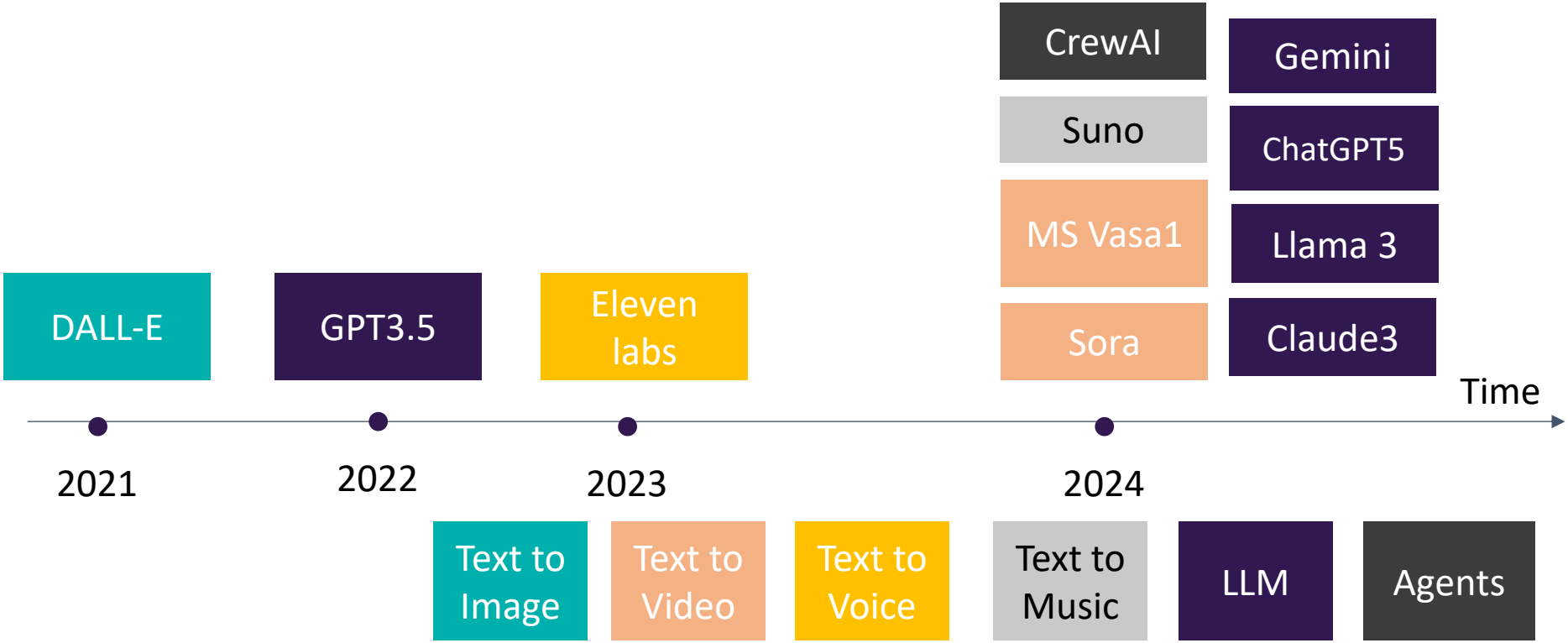
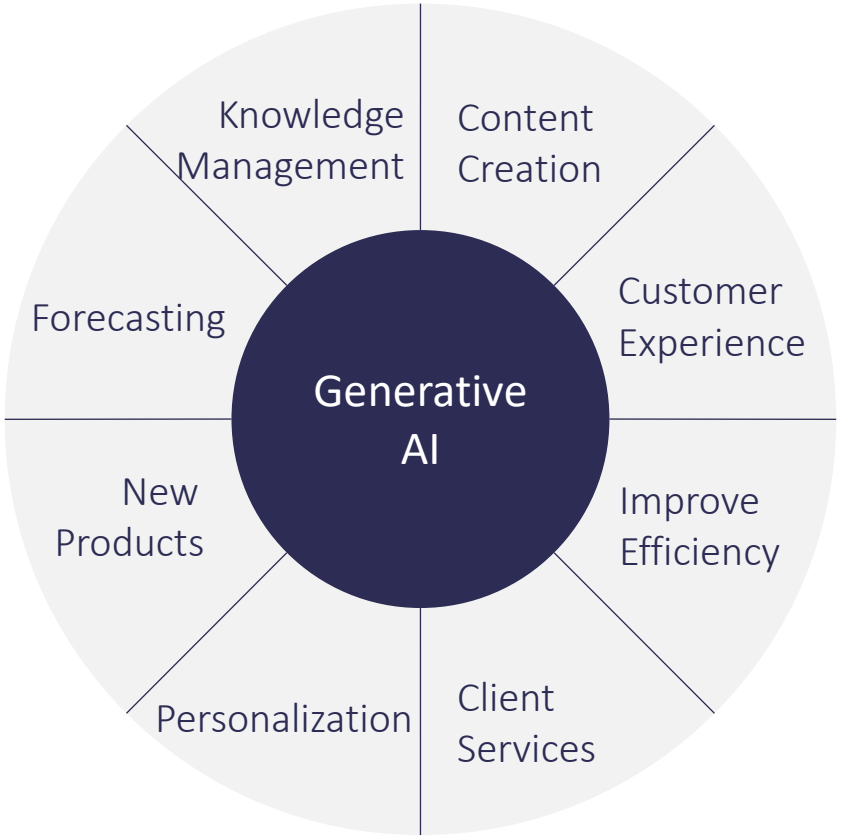
Recent Developments in AI-Tools



Generative AI

What is GenAI?

- Artificial Intelligence, that can create new content
- Trained on large datasets
- Typically based on Transformer architecture
- Applications
  - Text generation
  - Image/Music/Video generation
- Challenges
  - Ethics and Biases
  - Copyright/Ownership



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Which problems are ideal for AI?

Massive combinatorial space

Examples:
Chess
- 10^{64} combinations

Go
- 10^{172} combinations
- more than atoms in the universe

Clear objective function for optimization

- Recommer systems: max click-through rate
- Images: min classification error
- Games: max score
- Fraud Detection: min false negatives

Data

- many data or
- simulator that allows creation of many data

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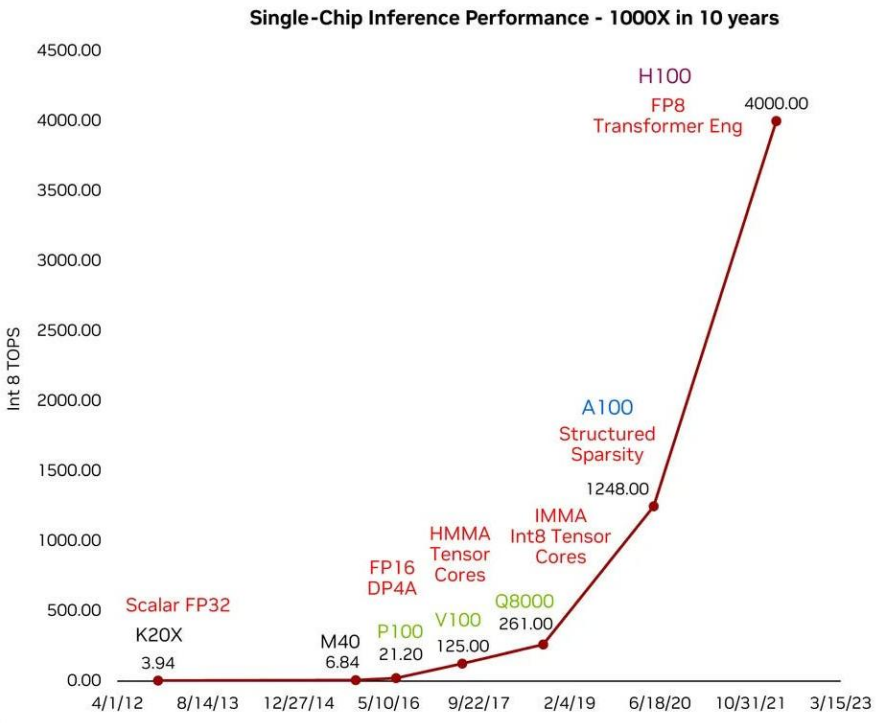
Larger models require more computational power

- Moore‘s law is still active in chip performance



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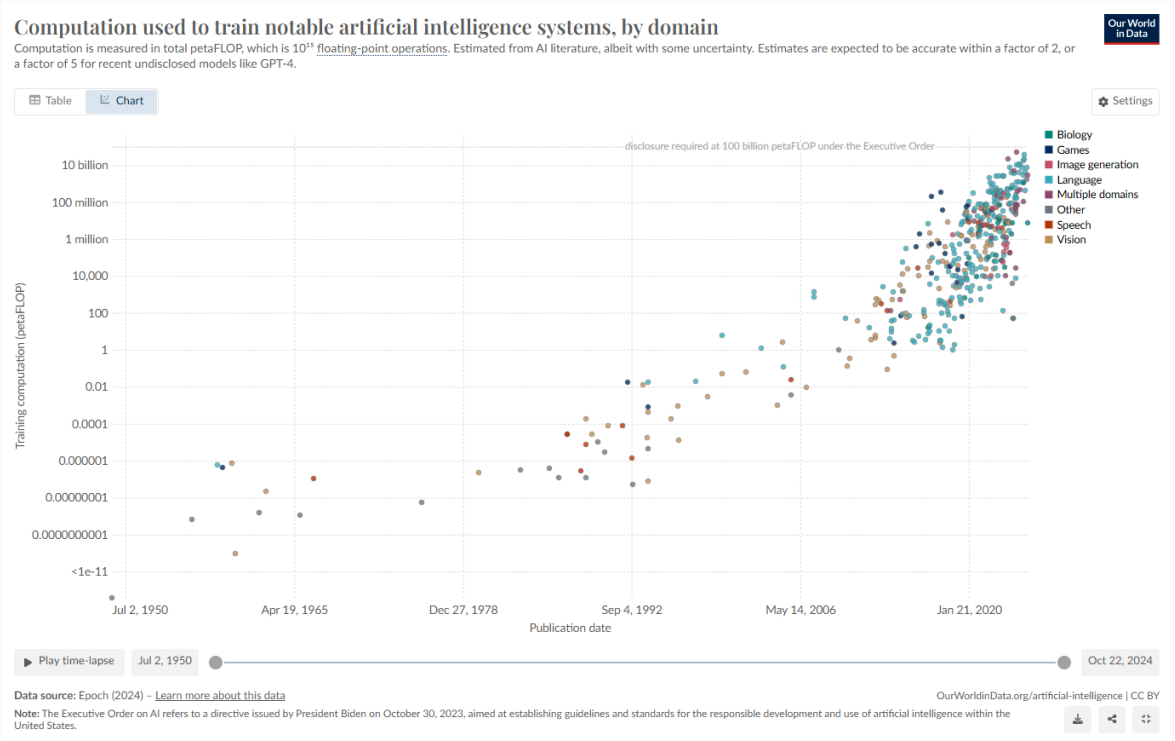

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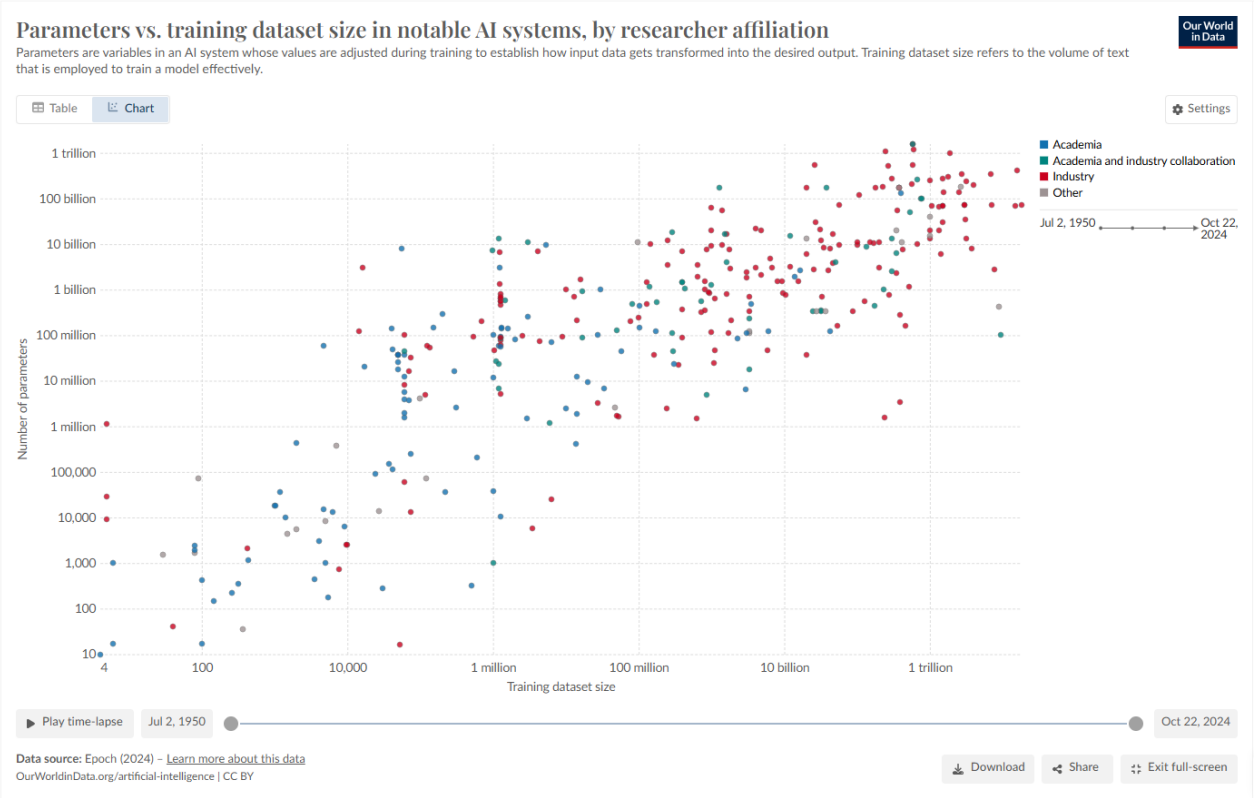
Model Sizes increase exponentially

- Based on larger computational power
- Larger datasets



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Investments

Data compiled May 2, 2024.
LLM = large language model; ROF = round of funding.
NA = not available.
Funding rounds between Jan. 31, 2023, and May 2, 2024, where target company topic tags or categories includes "artificial intelligence."
Excludes companies whose primary offering is not tied to AI.
Design credit: Zain Ullah.
Source: S&P Global Market Intelligence.
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Source: https://www.spglobal.com/marketintelligence/en/news-insights/latest-news-headlines/how-top-vcs-are-placing-their-growing-ai-bets-81722741

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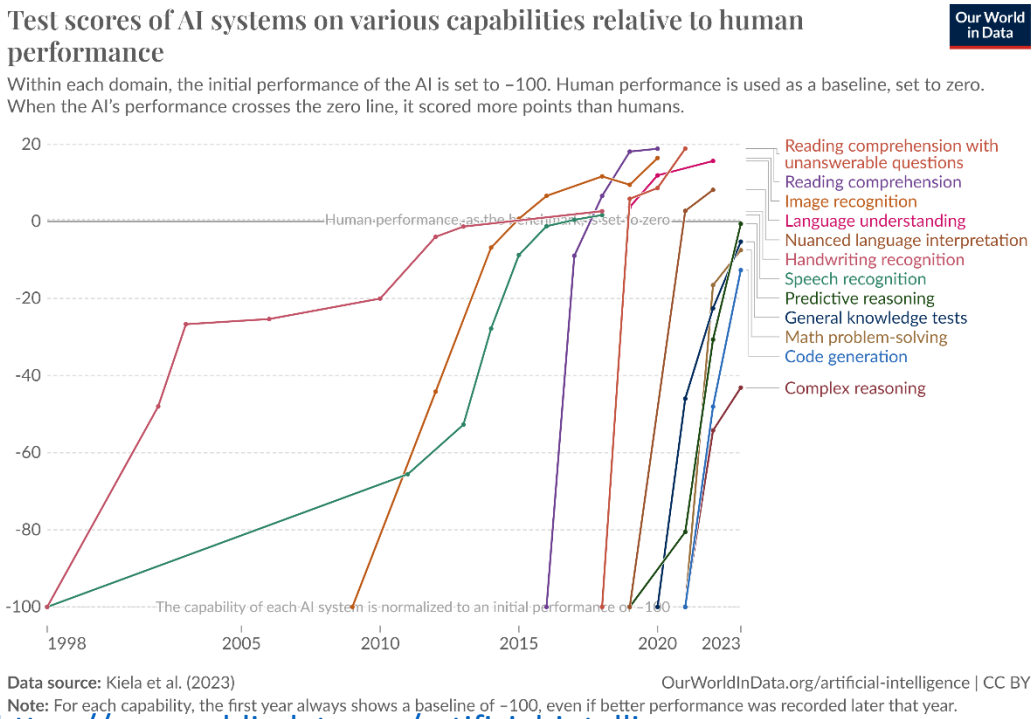
Current State: Exceeding human capabilities



Source: https://ourworldindata.org/artificial-intelligence

AI already exceeds human performance in nearly all tasks.

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Current State: LLMs get exponentially better

Rank* (UB)	Model	Arena Score	95% CI	Votes	Organization	License	Knowledge Cutoff
1	ChatGPT-4o-latest (2024-08-08)	1316	+4/-4	24358	OpenAI	Proprietary	2023/10
2	Gemini-1.5-Pro-Exp-0827	1301	+5/-5	19976	Google	Proprietary	2023/11
2	Gemini-1.5-Pro-Exp-0801	1298	+4/-3	25471	Google	Proprietary	2023/11
2	Grok-2-08-13	1295	+4/-6	10170	xAI	Proprietary	2024/3
5	GPT-4o-2024-05-13	1286	+3/-3	83181	OpenAI	Proprietary	2023/10
6	GPT-4o-mini-2024-07-18	1274	+4/-4	23318	OpenAI	Proprietary	2023/10
6	Gemini-1.5-Flash-Exp-0827	1270	+7/-6	6610	Google	Proprietary	2023/11
6	Claude 3.5 Sonnet	1270	+3/-3	53610	Anthropic	Proprietary	2024/4
6	Gemini Advanced App (2024-05-14)	1266	+3/-3	52225	Google	Proprietary	Online
6	Grok-2-Mini-08-13	1266	+6/-6	10939	xAI	Proprietary	2024/3
7	Meta-Llama-3.1-405b-Instruct	1266	+3/-4	24855	Meta	Llama 3.1 Community	2023/12
7	GPT-4o-2024-08-06	1262	+7/-5	14960	OpenAI	Proprietary	2023/10
10	Gemini-1.5-Pro-001	1259	+3/-3	74909	Google	Proprietary	2023/11

Source: https://lmarena.ai/?leaderboard

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Current State: Math - 07/2024: Silver Medal in International Mathematical Olympiad

February 9, 2024

TIPS AND TRICKS

Why is ChatGPT bad at even basic math?

Two things ChatGPT can get wrong: math and facts. Why?

How is one of the biggest AI Tools in the world making mistakes that even a normal person can figure out?

Sunny K. Tuladhar · Follow

4 min read · Feb 3, 2024

You

1+1+1+3+1+1+1 -3+2+1+2+3

‹ 2/2 ›

ChatGPT

Let's simplify the expression:

$$1+1+1+3+1+1+1-3+2+1+2+3.$$

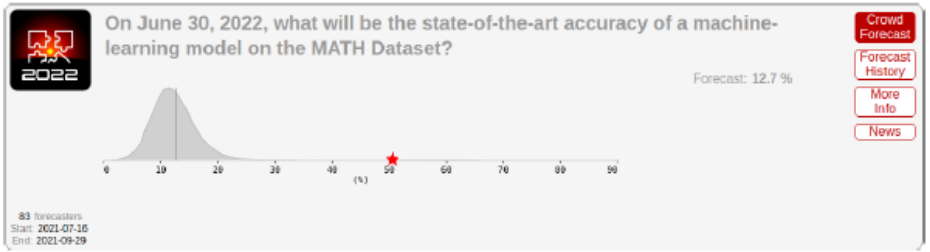
Combine the like terms:

$$1+1+1+3+1+1+1-3+2+1+2+3=18.$$

So, the result is 18.

The fact that the program can come up with a non-obvious construction like this is very impressive, and well beyond what I thought was state of the art.

PROF SIR TIMOTHY GOWERS,
IMO GOLD MEDALIST AND FIELDS MEDAL WINNER



Source: https://lmarena.ai/?leaderboard

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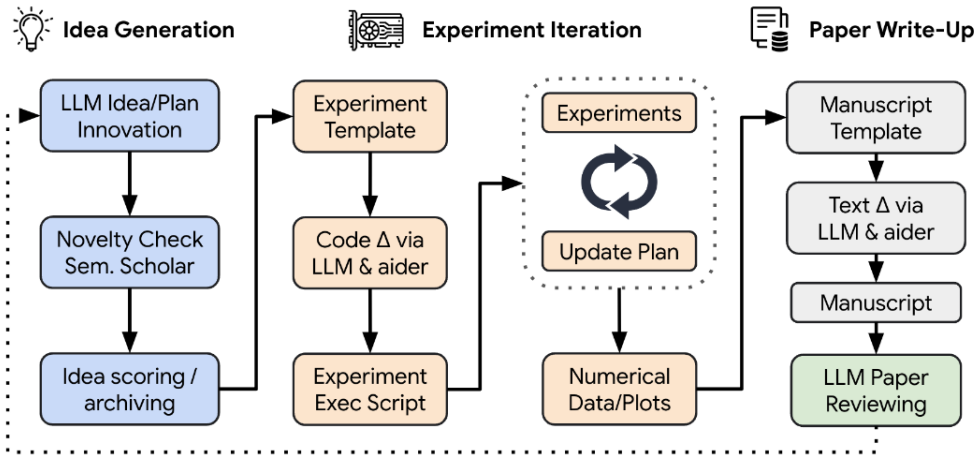
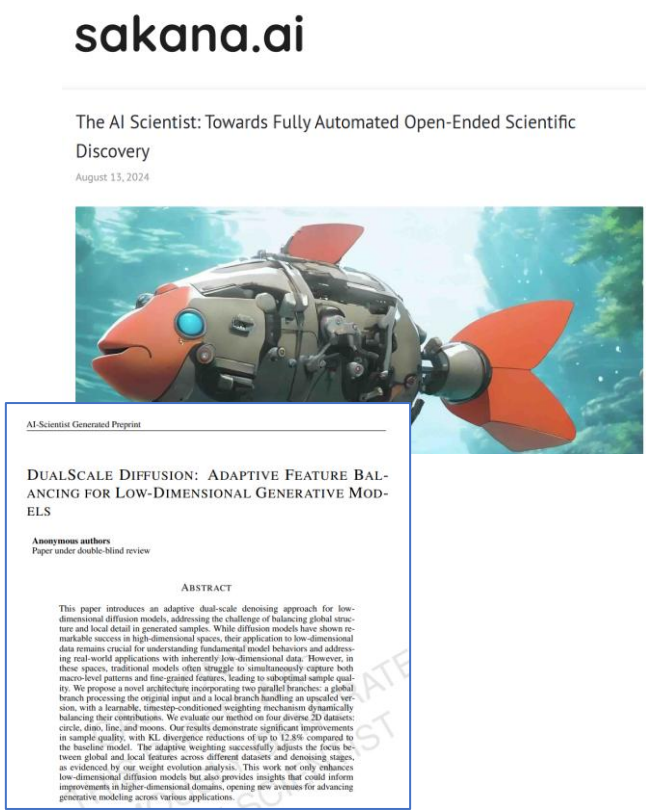
Current State: AI Scientist (2024-08)

- AI system can write research papers fully automated
- full-paper cost of $15





Source: https://sakana.ai/ai-scientist/

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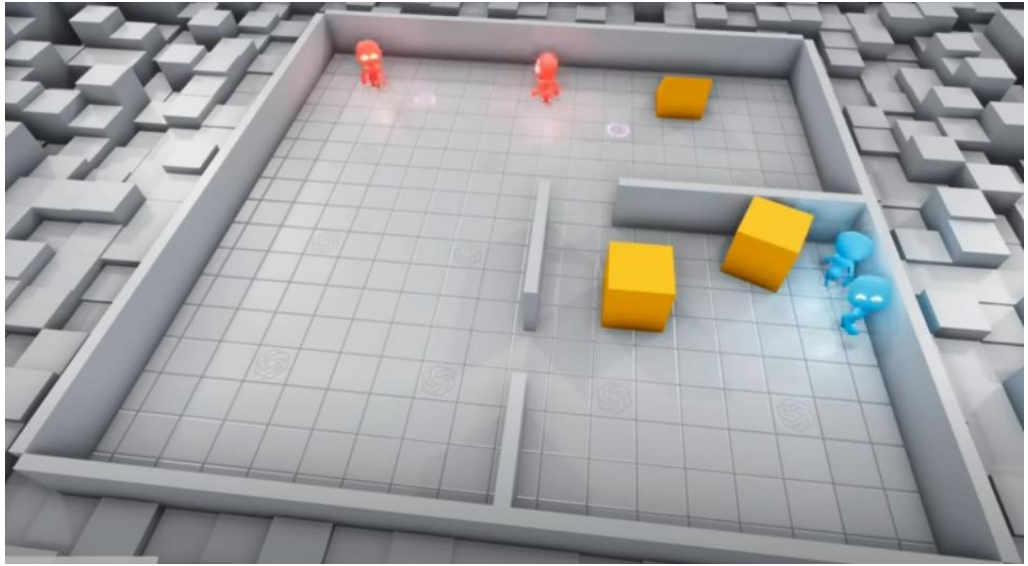
Current State: AI agents collaboration

- AI teams play hide-and-seek
- two hiders vs. two seekers
- develop novel solutions
- find glitches in programming
- unexpected behavior



Source: https://www.youtube.com/watch?v=Lu56xVlZ40M&t=5s

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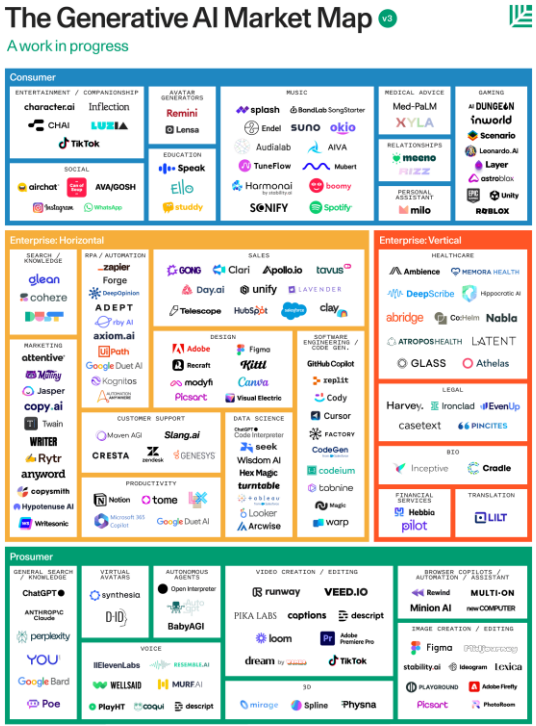
Generative AI Market



Source: https://www.sequoiacap.com/article/generative-ai-act-two/

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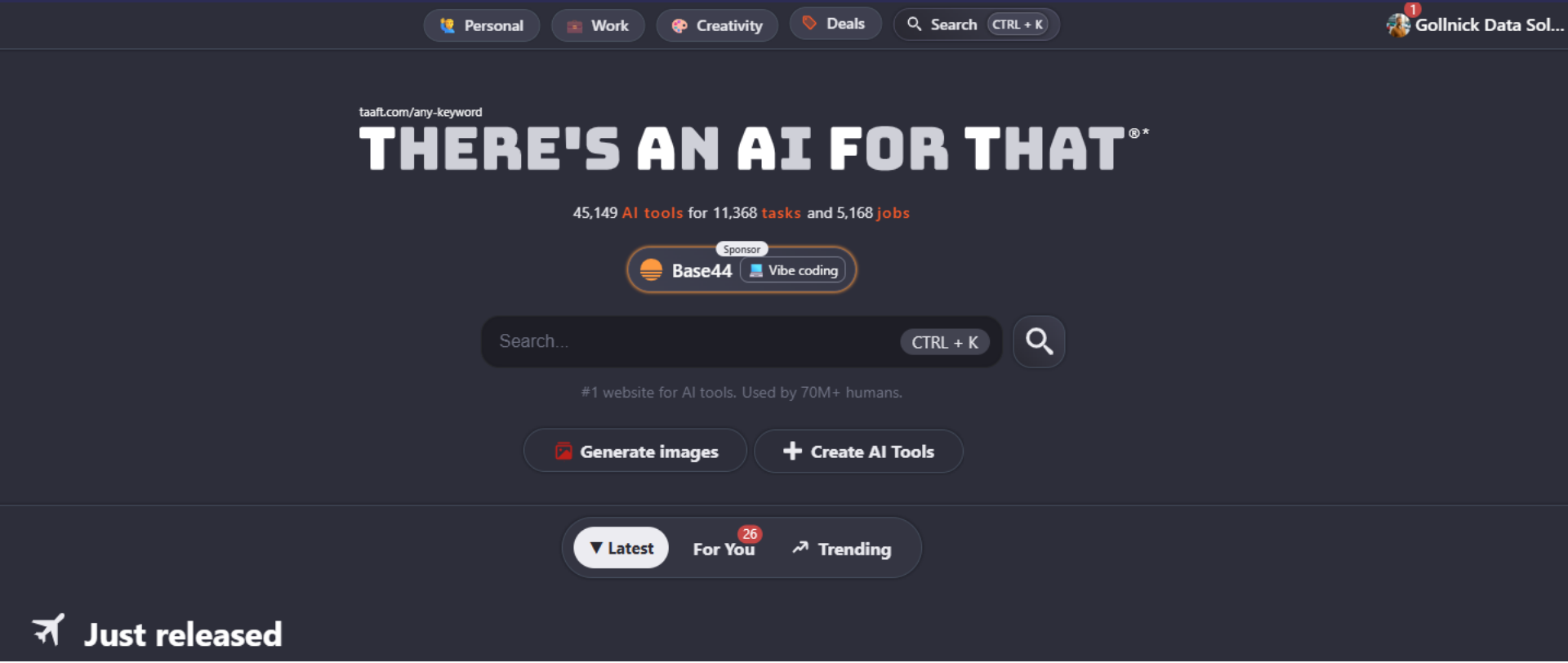
Generative AI Market



Source: https://theresanaiforthat.com/